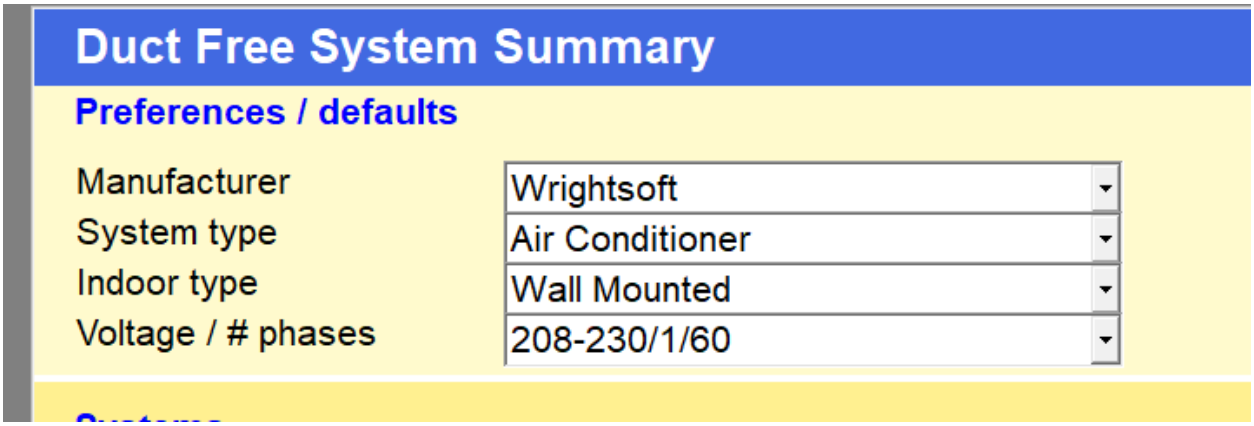
Mini-Split Manual

Mini-Split Systems Design Steps

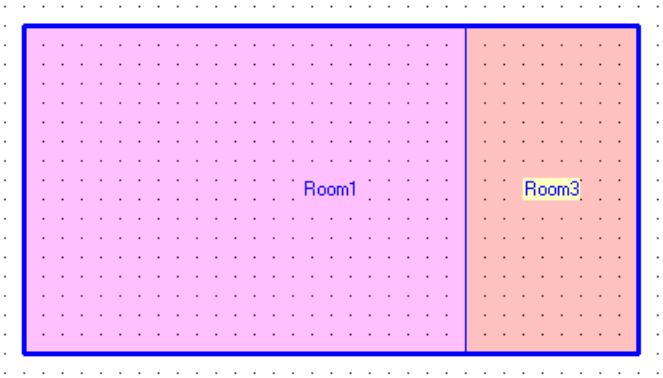
- In the Duct Free System Summary screen setup Mini-Split Manufacturer, System Type and Unit type preferences. Save those settings in the default template.
- Draw rooms, windows, doors, etc for the program to calculate room by room heating and cooling loads.
- Use Zone Tree to create Outdoor and Indoor Mini-Split zones, assign rooms to them. Right-Draw will automatically create Indoor, Outdoor, and refrigeration pipe objects based on Zone Tree Configurations. All mini-split system objects will be put on 'Duct Free' layer.
- Position indoor and outdoor unit where you want them to be. Use 'edit point' command to layout refrigeration piping. Modify mini-split objects properties as needed.
- Check calculation results in object properties, Duct Free System Summary screen.
- Check bill of materials, it will be automatically generated from your mini-split design.
- Print calculation reports: Duct Free category in the report list

Mini-Split System Design details

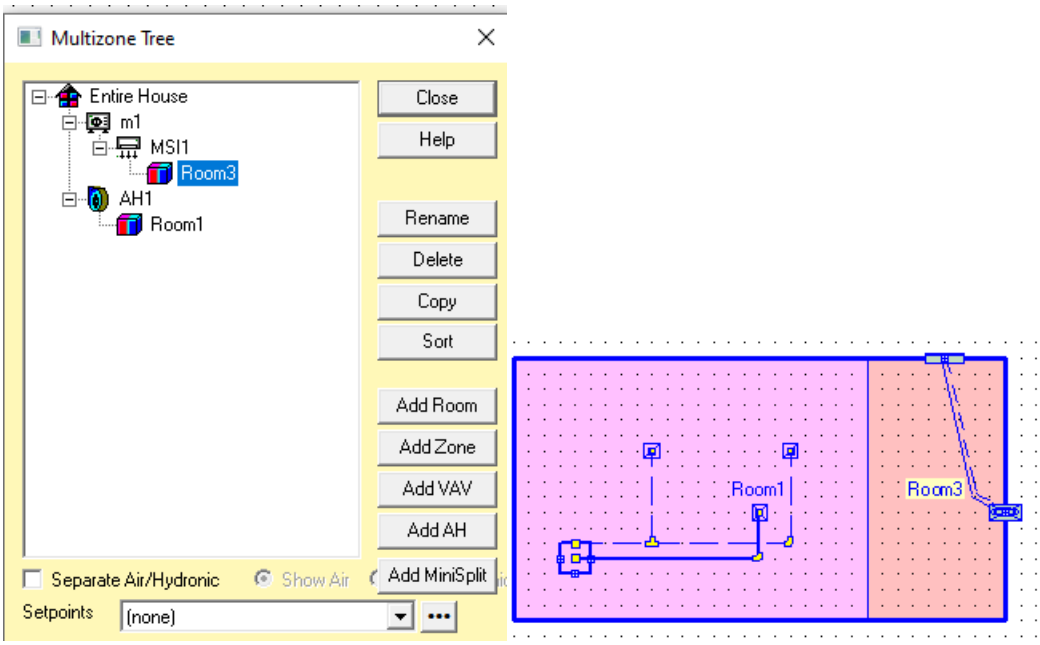
Select preferences for manufacturer, system type unit type and voltage in Duct Free System Summary screen.



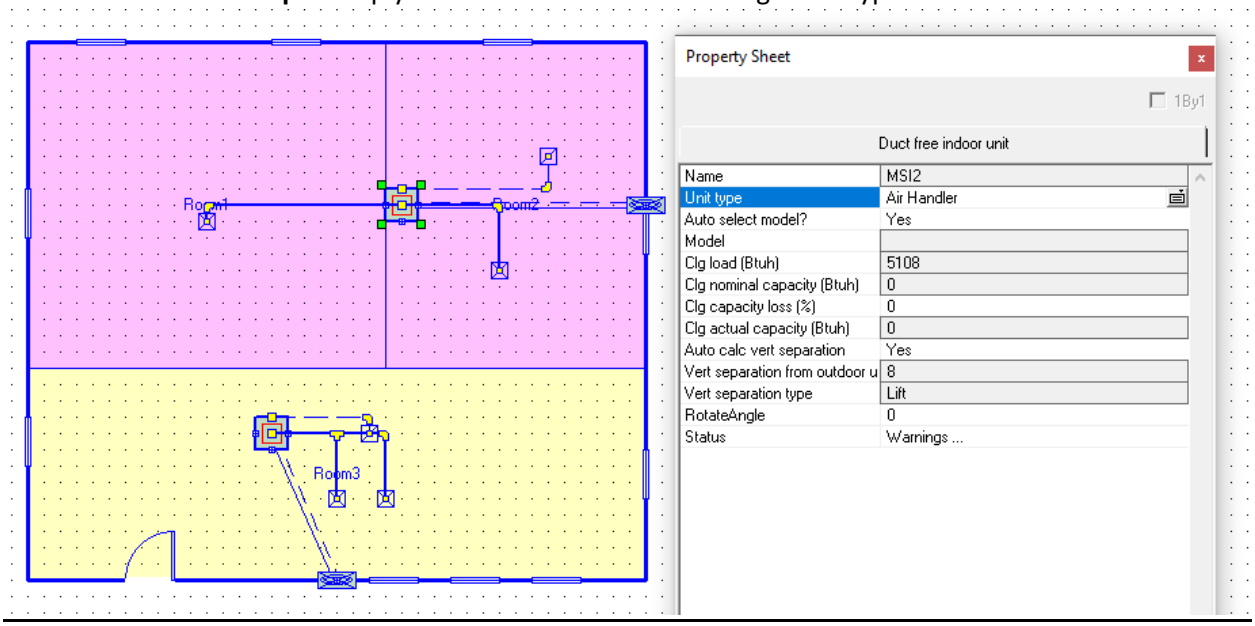
Create or open existing room by room design.



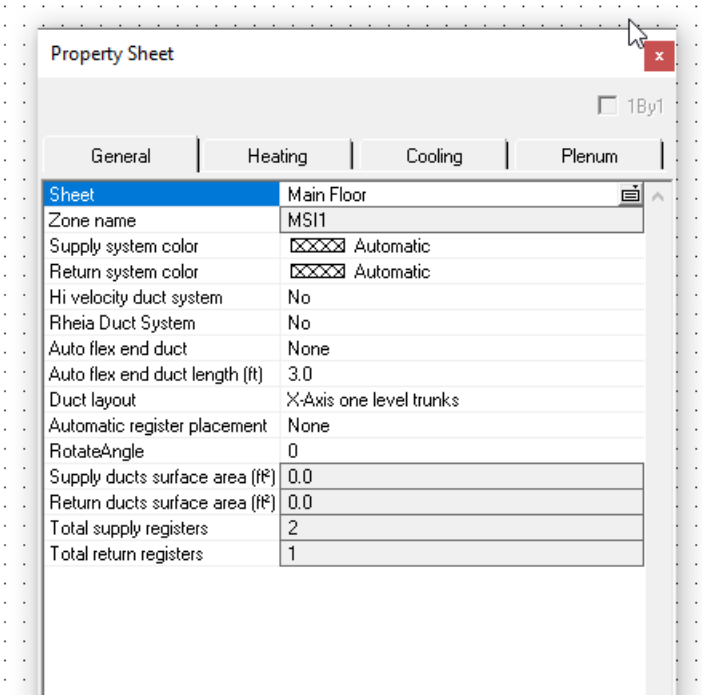
To add Mini-Split go to Multizone Tree to select **{ Add MiniSplit** } you can add multiple units or heads to rooms inside the Multizone Tree. To have a Mini-Split system with a Conventional system in the same building, you must have multiple rooms. In the Multizone Tree click the room you want a Mini-Split in and add. Like I Have in example here with both Conventional and Mini-Split.



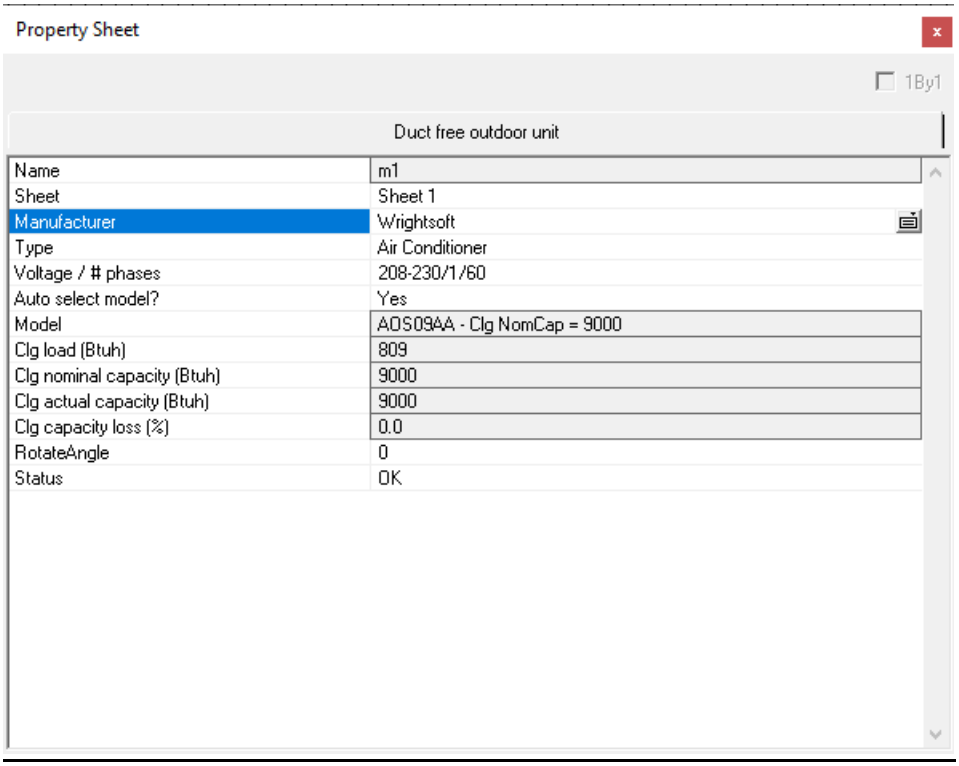
To make **Ducted Mini-split** simply click on indoor unit and change Unit type to ‘Air handler’ or ‘Ducted’.



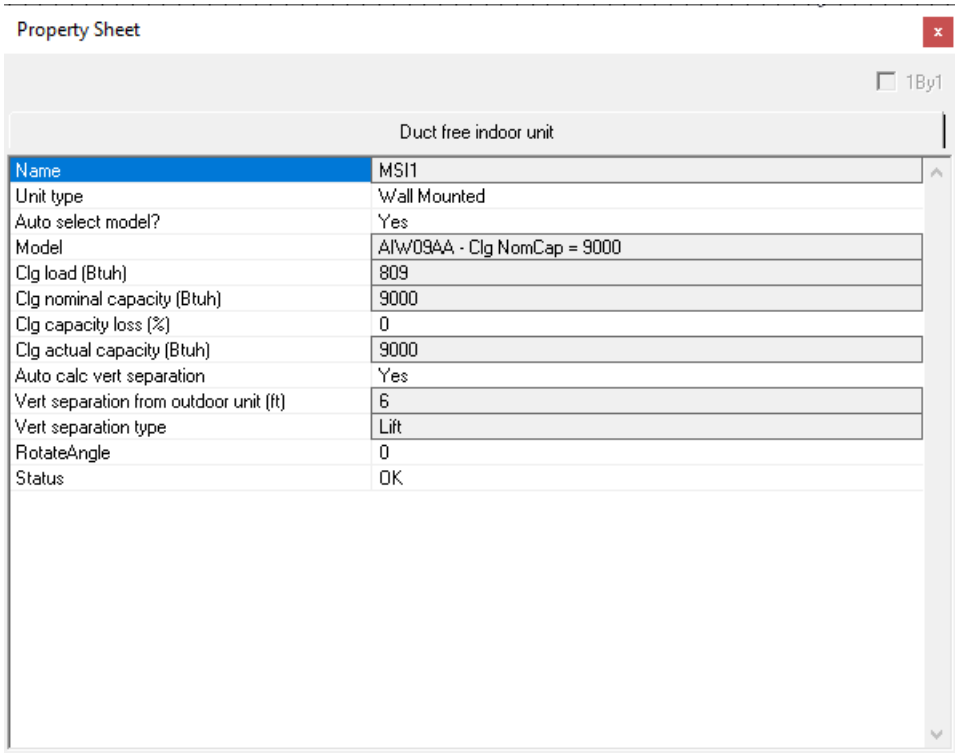
To make ducts appear right-click indoor unit and change duct layout to X axis.



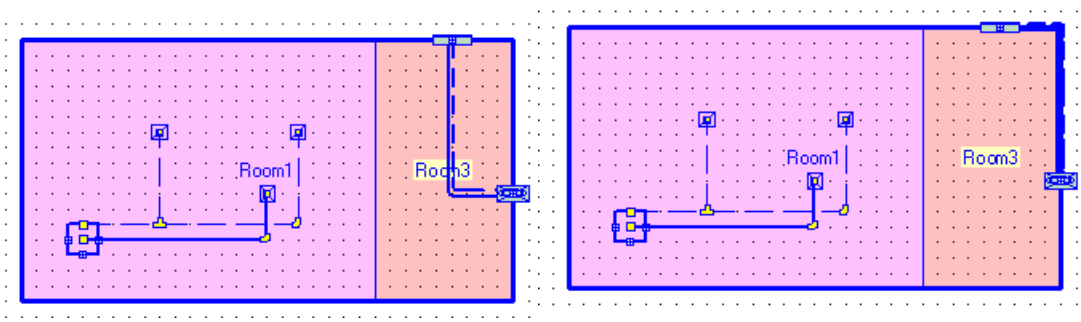
After adding your Mini-Split system to choose Manufacturer click on the Mini-Split Condenser and then **Right click** to bring up the properties sheet then choose the Manufacturer along with other properties.



When choosing Mount type click on Mini-Split head unit and then **right-click** to bring up the properties sheet then choose Unit type.



You can position indoor and outdoor units where you want in the building, you can also move the layout of the refrigeration lines by using the edit points command.



Modify Mini-Split system object properties as needed.

Property Sheet

1By1

Duct free outdoor unit

Name	m1
Sheet	Sheet 1
Manufacturer	Wrightsoft
Type	Air Conditioner
Voltage / # phases	208-230/1/60
Auto select model?	Yes
Model	AOS09AA - Clg NomCap = 9000
Clg load (Btuh)	809
Clg nominal capacity (Btuh)	9000
Clg actual capacity (Btuh)	9000
Clg capacity loss (%)	0.0
RotateAngle	0
Status	OK

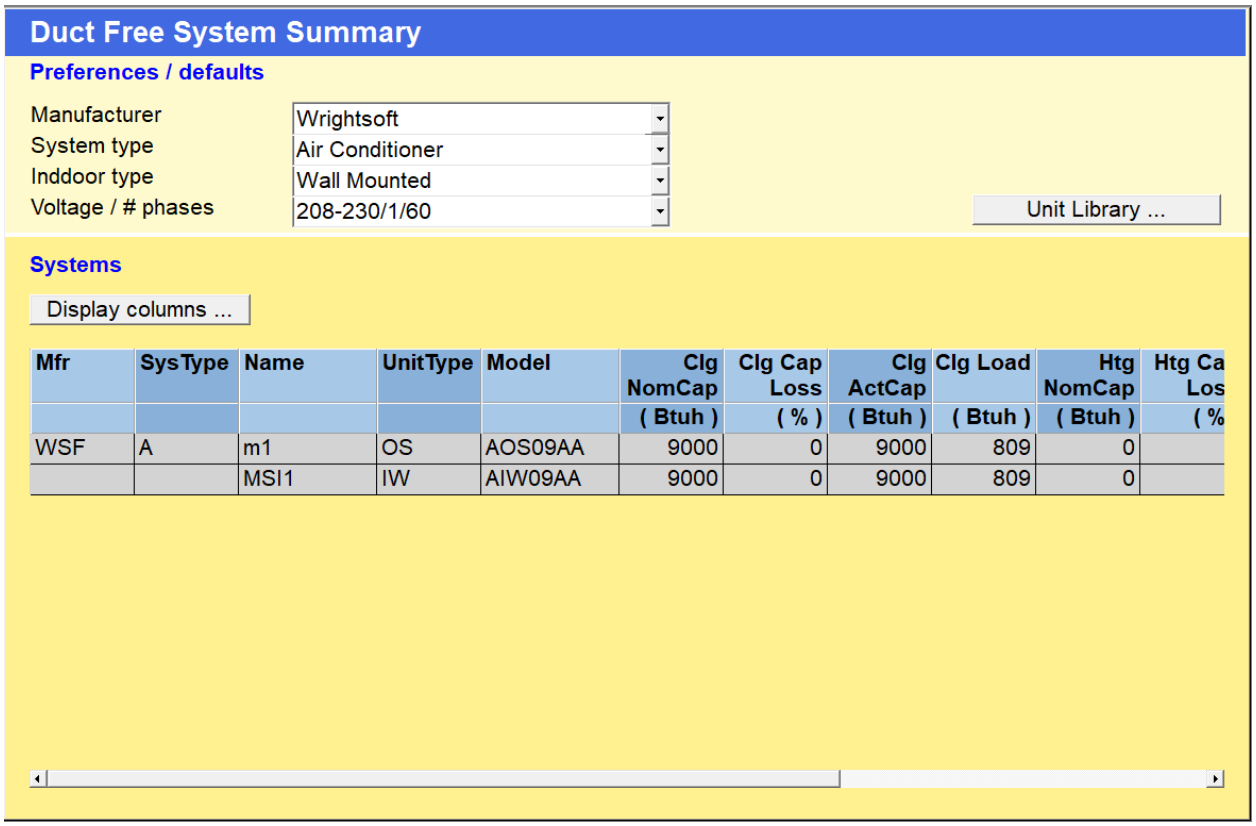
Property Sheet

1By1

Duct free indoor unit

Name	MSI1
Unit type	Wall Mounted
Auto select model?	Yes
Model	AIW09AA - Clg NomCap = 9000
Clg load (Btuh)	809
Clg nominal capacity (Btuh)	9000
Clg capacity loss (%)	0
Clg actual capacity (Btuh)	9000
Auto calc vert separation	Yes
Vert separation from outdoor unit (ft)	6
Vert separation type	Lift
RotateAngle	0
Status	OK

Check calculation results in the Duct Free System Summary screen



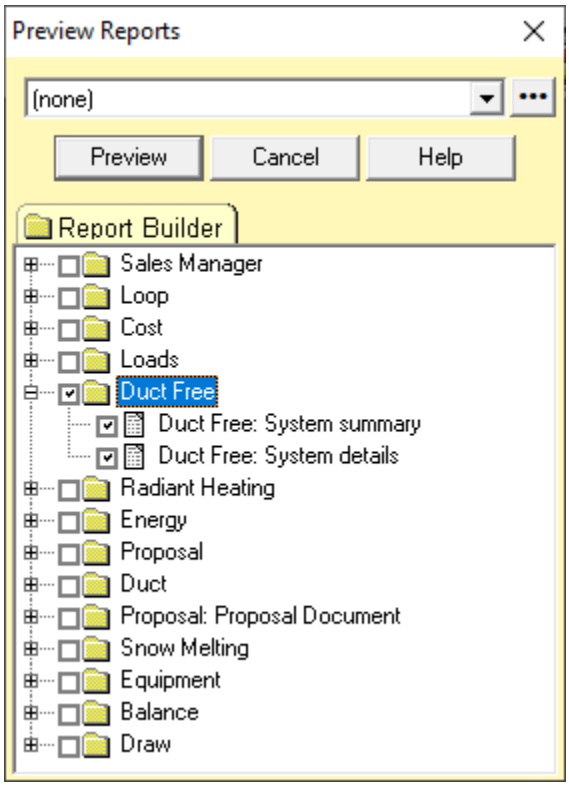
Mfr	SysType	Name	UnitType	Model	Clg NomCap	Clg Cap Loss	Clg ActCap	Clg Load	Htg NomCap	Htg Ca Los
					(Btuh)	(%)	(Btuh)	(Btuh)	(Btuh)	(%
WSF	A	m1	OS	AOS09AA	9000	0	9000	809	0	
		MSI1	IW	AIW09AA	9000	0	9000	809	0	

To check Bill Of Materials which is automatically generated go to (**Show** then **Right-Proposal** then **Bill Of Materials**)

Bill of Materials Base System Split AC + Furn Filter... Locked

Src	Name	Description	Phase	Qty	Un	Tax	Price	Ext	Cost	Ext cost
		Materials								
		Subtotal, Materials						0.00		0.00
		Labor								
		Subtotal, Labor						0.00		0.00
		Subcontractor								
		Subtotal, Subcontractor						0.00		0.00
		Accessories								
		Subtotal, Accessories						0.00		0.00
		Miscellaneous								
		Subtotal, Miscellaneous						0.00		0.00
		Mini Split System								
WSF	AOS12AA	Single Split AC 12000 btuh	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AIA12AA	MiniSplit AC Indoor Air Handler Unit 120(	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AOS09AA	Single Split AC 9000 btuh	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AIW09AA	MiniSplit AC Indoor Wall Mounted Unit 9(	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AOM18AA	Multi Split AC 18000 btuh	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AIC09AA	MiniSplit AC Indoor Ceiling Cassette Uni	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	AIW12AA	MiniSplit AC Indoor Wall Mounted Unit 1	None	1.00	ea	☐	0.00	0.00	0.00	0.00
WSF	RP0375	3/8 inch Refrigeration pipe	None	77.00	ea	☐	0.00	0.00	0.00	0.00
WSF	RP0250	1/4 inch Refrigeration pipe	None	77.00	ea	☐	0.00	0.00	0.00	0.00
WSF	ELS0375	3/8 inch 90 deg elbow	None	17.00	ea	☐	0.00	0.00	0.00	0.00
WSF	ELS0250	1/4 inch 90 deg elbow	None	17.00	ea	☐	0.00	0.00	0.00	0.00
		Subtotal, Mini Split System						0.00		0.00
		Overall total						**377.20**		**377.20**

Then print calculation reports- Duct free category in the report.



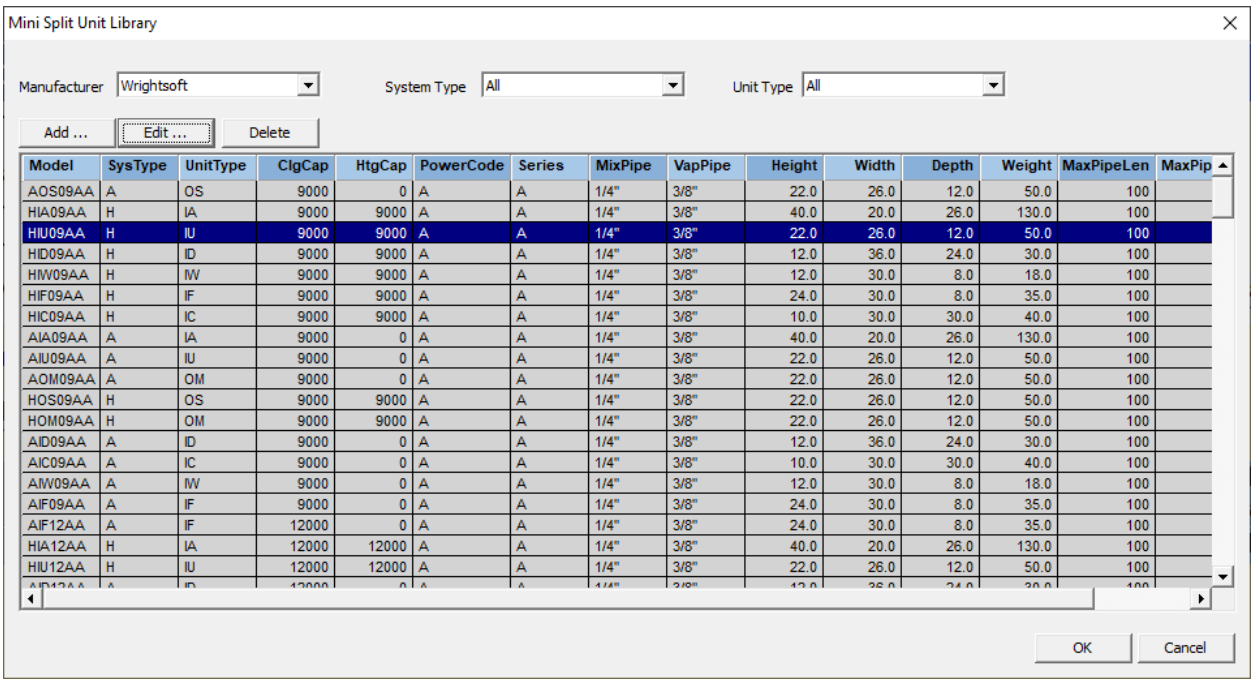
Mini-Split Unit and Parts Database

Mini-Split Unit data and parts are included in the program as a part of Right-Proposal database.

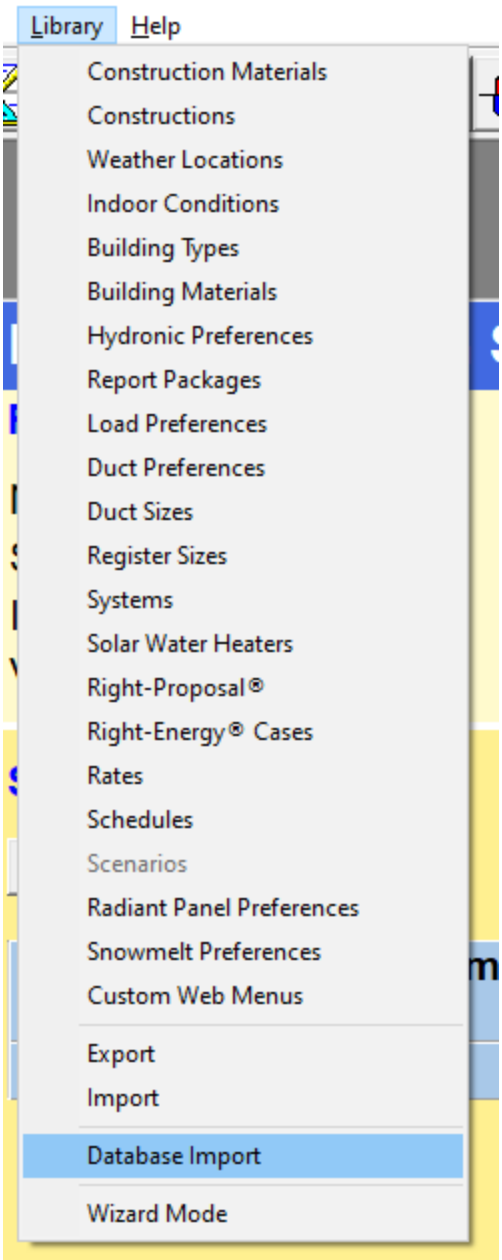
Users can see available units per manufacturer in the ‘Mini-Split Unit Library’ screen. You can access this screen by going to the Duct Free System Summary screen and pressing ‘Unit Library’ button.

Users can add / modify Mini-Split unit data by

a) Using UI in ‘Mini-Split Unit Library’ screen
b) Using Database Import from excel



Model	SysType	UnitType	ClgCap	HtgCap	PowerCode	Series	MixPipe	VapPipe	Height	Width	Depth	Weight	MaxPipeLen	MaxPip
AOS09AA	A	OS	9000	0	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
HIA09AA	H	IA	9000	9000	A	A	1/4"	3/8"	40.0	20.0	26.0	130.0	100	
HIU09AA	H	IU	9000	9000	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
HID09AA	H	ID	9000	9000	A	A	1/4"	3/8"	12.0	36.0	24.0	30.0	100	
HIW09AA	H	IW	9000	9000	A	A	1/4"	3/8"	12.0	30.0	8.0	18.0	100	
HIF09AA	H	IF	9000	9000	A	A	1/4"	3/8"	24.0	30.0	8.0	35.0	100	
HIC09AA	H	IC	9000	9000	A	A	1/4"	3/8"	10.0	30.0	30.0	40.0	100	
AIA09AA	A	IA	9000	0	A	A	1/4"	3/8"	40.0	20.0	26.0	130.0	100	
AIU09AA	A	IU	9000	0	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
AOM09AA	A	OM	9000	0	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
HOS09AA	H	OS	9000	9000	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
HOM09AA	H	OM	9000	9000	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	
AID09AA	A	ID	9000	0	A	A	1/4"	3/8"	12.0	36.0	24.0	30.0	100	
AIC09AA	A	IC	9000	0	A	A	1/4"	3/8"	10.0	30.0	30.0	40.0	100	
AIW09AA	A	IW	9000	0	A	A	1/4"	3/8"	12.0	30.0	8.0	18.0	100	
AIF09AA	A	IF	9000	0	A	A	1/4"	3/8"	24.0	30.0	8.0	35.0	100	
AIF12AA	A	IF	12000	0	A	A	1/4"	3/8"	24.0	30.0	8.0	35.0	100	
HIA12AA	H	IA	12000	12000	A	A	1/4"	3/8"	40.0	20.0	26.0	130.0	100	
HIU12AA	H	IU	12000	12000	A	A	1/4"	3/8"	22.0	26.0	12.0	50.0	100	


Library
Help
Construction Materials
Constructions
Weather Locations
Indoor Conditions
Building Types
Building Materials
Hydronic Preferences
Report Packages
Load Preferences
Duct Preferences
Duct Sizes
Register Sizes
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Right-Energy® Cases
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Schedules
Scenarios
Radiant Panel Preferences
Snowmelt Preferences
Custom Web Menus
Export
Import
Database Import
Wizard Mode

Database Import

Choosing What To Update

The Excel Import mechanism supports the import of parts and equipment data. If equipment and parts data are both to be imported the equipment must be represented in the parts data using the same manufacturer code and part number or they will be ommitted from the import.

Please select which data you are importing from the choices below

- [x] Include Right-Proposal® parts data
- [ ] Include equipment data

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